ALICE L WALTON SCHOOL OF MEDICINE

Epidemiology of Cancer

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Adapted from: Epidemiology of Cancer

Listen to this Brick
0:00 / 30:37 1x

Learning Objectives (3) Versions

After completing this brick, you will be able to:

1. Explain the epidemiology of cancer and how cancer incidence has changed over time
2. Discuss key risk factors for common cancers
3. Formulate strategies to reduce the risk of cancer

CASE CONNECTION

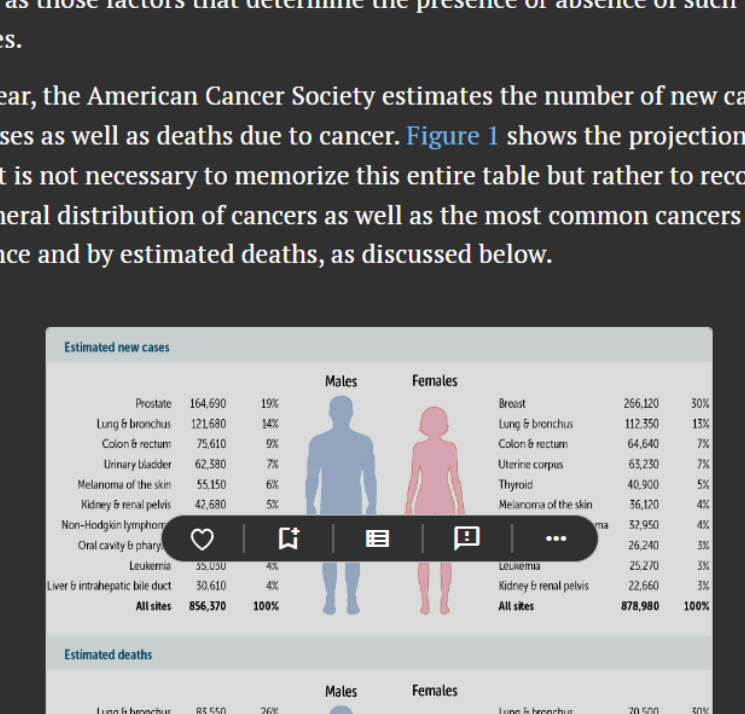
A 62-year-old male who is referred to you by his dentist for a 2.0 cm, shallow ulcer of the mouth. The patient reports that he's had the lesion for months and that it doesn't bother him except when he eats something acidic. He has no family history of cancer. You consider what risk factors may have contributed to the lesion and what the next step in evaluation should be.

GO TO CONCLUSION ↓

What Kinds of Cancer Are Most Common? (LO#1)

Cancer epidemiology is the study of the distribution of cancers in a population as well as those factors that determine the presence or absence of such diseases.

Each year, the American Cancer Society estimates the number of new cancer diagnoses as well as deaths due to cancer. Figure 1 shows the projections for 2018. It is not necessary to memorize this entire table but rather to recognize the general distribution of cancers as well as the most common cancers by incidence and by estimated deaths, as discussed below.



Estimated new cases

Males			Females		
Prostate	164,690	19%	Breast	266,120	30%
Lung & bronchus	121,680	14%	Lung & bronchus	112,350	13%
Colon & rectum	75,610	9%	Colon & rectum	64,640	7%
Urinary bladder	62,380	7%	Uterine corpus	63,230	7%
Melanoma of the skin	55,150	6%	Thyroid	40,900	5%
Kidney & renal pelvis	42,680	5%	Melanoma of the skin	36,120	4%
Non-Hodgkin lymphoma				32,950	4%
Oral cavity & pharynx				26,240	3%
Leukemia	35,030	4%	Leukemia	25,270	3%
Liver & intrahepatic bile duct	30,610	4%	Kidney & renal pelvis	22,660	3%
All sites	856,370	100%	All sites	878,980	100%

Estimated deaths

Males			Females		
Lung & bronchus	83,550	26%	Lung & bronchus	70,500	[illegible]
Prostate	29,430	9%	Breast	40,920	[illegible]
Colon & rectum	27,390	8%	Colon & rectum	23,240	7%
Pancreas	23,020	7%	Pancreas	21,310	7%
Liver & intrahepatic bile duct	20,540	6%	Ovary	14,070	5%
Leukemia	14,270	4%	Uterine corpus	11,350	4%
Esophagus	12,850	4%	Leukemia	10,100	4%
Urinary bladder	12,520	4%	Liver & intrahepatic bile duct	9,660	3%
Non-Hodgkin lymphoma	11,510	4%	Non-Hodgkin lymphoma	8,400	3%
Kidney & renal pelvis	10,010	3%	Brain & other nervous system	7,340	3%
All sites	323,630	100%	All sites	286,010	100%

Figure 1

There are many different types of cancer, and each has its own prognosis. Some types of cancer, like thyroid cancer, are almost always curable. Others, like pancreatic cancer, are almost always rapidly fatal. This is why the lists of the most common cancers and the cancers causing the most deaths look different.

For example, take a look at the most common cancers in men (prostate) and women (breast) in Figure 1. Although these cancers can be fatal, their overall prognosis is relatively good compared with lung cancer, which has a poor prognosis. That's why so many more women die of lung cancer than breast cancer, even though the incidence of breast cancer is more than double the incidence of lung cancer. Thyroid cancer is fairly common, especially in women, but its prognosis is so good that it doesn't even make the list of cancers causing the most deaths in women. Pancreatic cancer is uncommon, but it causes almost as many deaths as does colon cancer.

ACTIVE LEARNING QUESTION

What are the most common cancers in the United States?

A. ☑ Breast (female)
B. ☑ Colon/rectum
C. ☑ Lung
D. ☐ Pancreas
E. ☑ Prostate (male)

Explanation

The most common cancers in the United States are those of the breast (female), colon/rectum, lung, and prostate (male). Notably, lung cancer causes the most deaths in men and women in the United States. Pancreatic cancer, though uncommon, causes many deaths due to its aggressive behavior.

How Have Cervical, Colorectal, and Breast Cancers Changed in Incidence? (LO#1, LO#2, LO#3)

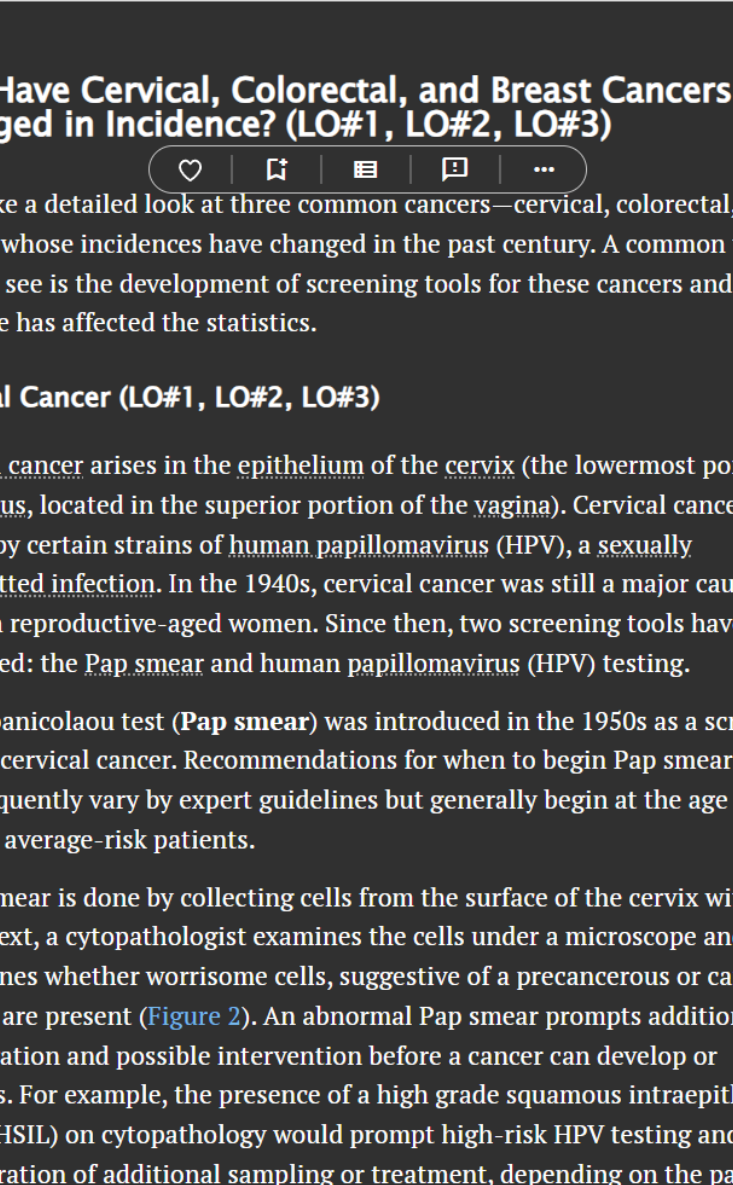
Let's take a detailed look at three common cancers—cervical, colorectal, and breast—whose incidences have changed in the past century. A common theme you will see is the development of screening tools for these cancers and how their use has affected the statistics.

Cervical Cancer (LO#1, LO#2, LO#3)

Cervical cancer arises in the epithelium of the cervix (the lowermost portion of the uterus, located in the superior portion of the vagina). Cervical cancer is caused by certain strains of human papillomavirus (HPV), a sexually transmitted infection. In the 1940s, cervical cancer was still a major cause of death in reproductive-aged women. Since then, two screening tools have been developed: the Pap smear and human papillomavirus (HPV) testing.

The Papanicolaou test (**Pap smear**) was introduced in the 1950s as a screening tool for cervical cancer. Recommendations for when to begin Pap smears and how frequently vary by expert guidelines but generally begin at the age of 21 years in average-risk patients.

A Pap smear is done by collecting cells from the surface of the cervix with a swab. Next, a cytopathologist examines the cells under a microscope and determines whether worrisome cells, suggestive of a precancerous or cancerous lesions, are present (Figure 2). An abnormal Pap smear prompts additional investigation and possible intervention before a cancer can develop or progress. For example, the presence of a high grade squamous intraepithelial lesion (HSIL) on cytopathology would prompt high-risk HPV testing and consideration of additional sampling or treatment, depending on the patient's age and risk-status.



Adapted from PathologyOutlines.com, credit Yevgen Chornenkyy, M.D., M.Sc. and Bonnie Choy, M.D.

Figure 2: Compared to a normal squamous cell (arrow), HSIL cells show increased nuclear-to-cytoplasmic ratios, nuclear enlargement, and nuclear membrane irregularity and raise concern for a pre-cancerous lesion.

The results have been dramatic. From 1955 to 1992 in the United States, both incidence and mortality from cervical cancer have decreased more than 60%! According to the most recent National Institutes of Health data, only about 7 in 100,000 US women developed cervical cancer in 2014; that's roughly 0.007% of the US female population. Worldwide, however, cervical cancer is still a scourge. In fact, 80% of cervical cancer occurs in developing countries. This is an area of global health that can be definitively addressed by increasing access to Pap smears in these countries.

HPV testing by PCR (polymerase chain reaction) is a more recently developed screening test for cervical cancer. Cervical cells are obtained in the same way as for a Pap smear and then are tested for high-risk strains of HPV. If high-risk HPV strains are present, the cervix is evaluated further for the presence of precancerous lesions. HPV PCR is being used more and may replace Pap smears in the future, at least in countries with access to this technology.

Colorectal Cancer (LO#1, LO#2, LO#3)

Colorectal cancer arises in the colon and rectum. It mostly occurs in people older than 50 years, although it can occur in younger people as well. The incidence of colorectal cancer has actually increased worldwide since the 1900s in parallel with economic development and longer life expectancy. This trend of increasing rates has been especially true in developing countries such as urban China, where Western lifestyles and diets have been adopted.

As of 2010, 5%-6% of individuals worldwide are expected to develop colorectal cancer in their lifetimes. From 2003 to 2012, however, the incidence of colorectal cancer in the United States has decreased significantly—by 2%-4% in all major ethnic groups and in both men and women. This may be due to increased awareness of colorectal cancer screening, aided by campaigns such as Screen for Life, a national project of the Centers for Disease Control and Prevention (CDC) in the United States. There are two tests to be familiar with: the fecal occult blood test and colonoscopy.

The presence of blood in the stool in any adult is suspicious for occult bleeding from a tumor in the lower gastrointestinal tract. This is the basis for the **fecal occult blood test**, a simple, cheap method of detecting blood in the stool. However, because blood in the stool is typically only present in more advanced colon cancers (and n on polyps), the utility of this test in detecting early cancer is limited.

The gold-standard screening test for colon cancer is **colonoscopy**, in which a scope is used to visualize the colon and remove any polyps (Figure 3) or other suspicious lesions before they have a chance to turn into colon cancer.

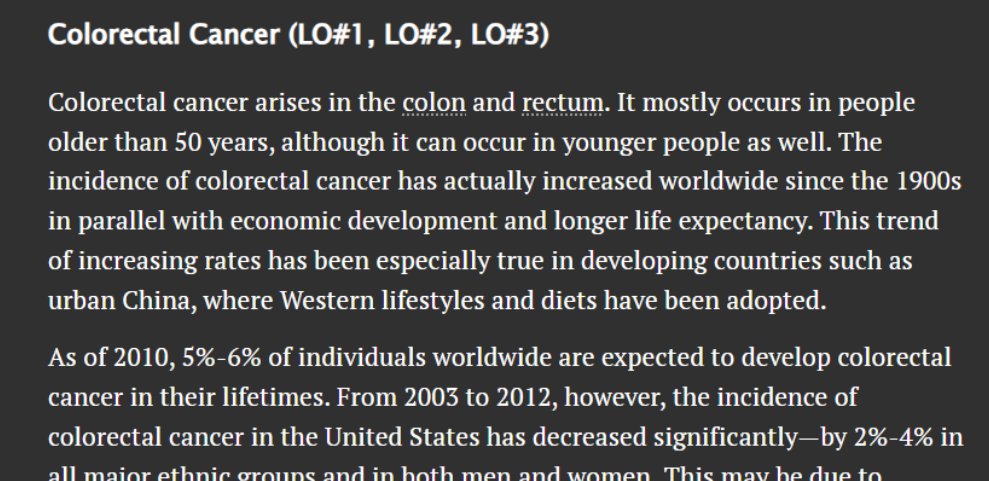


Figure 3: Colon adenomas, including a pedunculated example (A) and another with a velvety surface (B) can be removed during colonoscopy before the neoplasms progress to adenocarcinoma. Crowded, neoplastic glands are seen microscopically in the absence of invasion beyond the basement membrane and into the lamina propria (C).

Screening beginning at age 45 years is recommended for healthy individuals without a family history of cancer or personal history of cancer syndromes and is repeated every 10 years.

However, it is important to know that screening is based on individualized risk. For example, patients with a family history of sporadic colorectal adenocarcinoma sho years before the onset of the cancer in the family member.

What is the general trend (increasing or decreasing) of colorectal cancer in developing countries?

Breast Cancer (LO#1, LO#2, LO#3)

Breast cancer arises in the ducts and lobules of the breast. It is more prevalent in women than men, and the current lifetime risk of breast cancer in a woman is 1 in 8, making it the most frequently diagnosed cancer in women. The incidence of breast cancer increased in the latter part of the 20th century. One reason for the increase is the introduction of mammography.

Mammography, an X-ray of breast tissue to screen for breast cancer, first started being used widely around the 1970s-1980s. Mammograms have actually increased the apparent incidence of breast cancer simply because it allows us to detect it much sooner.

CLINICAL CORRELATION

Mammography screening for breast cancer is used to detect suspicious breast lesions, such as those showing microcalcifications, that need to undergo biopsy for future evaluation. The goal is to identify and treat developing cancers, such as ductal carcinoma in situ (Figure 4), before they progress to invasive or metastatic disease. For women with average risk of breast cancer, mammography screening usually begins at the age of 40 years.

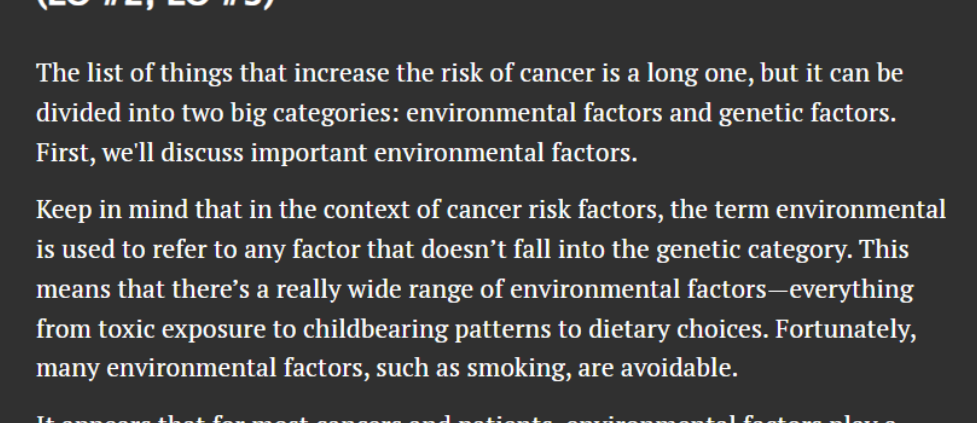


Figure 4: Microcalcifications on mammography are a suspicious finding for early breast cancer (left). Biopsy of the lesion showed neoplastic cells expanding a duct without invasion past the basement membrane (red arrowhead), consistent with ductal carcinoma in situ (right). Note the chunky, purple microcalcifications in the neoplastic lesion (right, square)

Estrogen is an important risk factor for luminal breast cancer due to stimulation of growt epithelial proliferation. Risk factors related to increased estrogen exposure include onset of menarche before the age of 12, delayed first live childbirth until after 30 or not having children, therapeutic or supplemental estrogen, and postmenopausal obesity, the latter of which is thought to be related to increased estrogen exposure.

What Are the Environmental Risk Factors for Cancer? (LO #2, LO #3)

The list of things that increase the risk of cancer is a long one, but it can be divided into two big categories: environmental factors and genetic factors. First, we'll discuss important environmental factors.

Keep in mind that in the context of cancer risk factors, the term environmental is used to refer to any factor that doesn't fall into the genetic category. This means that there's a really wide range of environmental factors—everything from toxic exposure to childbearing patterns to dietary choices. Fortunately, many environmental factors, such as smoking, are avoidable.

It appears that for most cancers and patients, environmental factors play a much bigger role than genetic factors. Figure 5 shows the estimated number of cancer deaths caused by identifiable factors. Note how small the family history slice is.

Much of our understanding in this area comes from epidemiologic studies showing that specific cancers are much more common in some parts of the world than in others. For example, breast cancer is much more common in the United States than in Japan. Further, when people emigrate from Japan to the United States, their ri ases—and in subsequent generations, the risk approaches that of native-born US citizens.

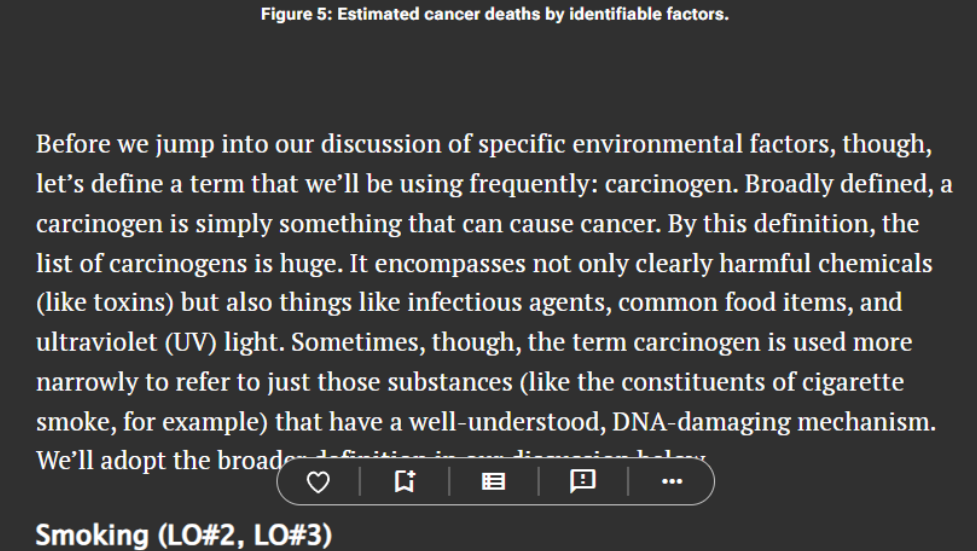


Figure 5: Estimated cancer deaths by identifiable factors.

Before we jump into our discussion of specific environmental factors, though, let's define a term that we'll be using frequently: carcinogen. Broadly defined, a carcinogen is simply something that can cause cancer. By this definition, the list of carcinogens is huge. It encompasses not only clearly harmful chemicals (like toxins) but also things like infectious agents, common food items, and ultraviolet (UV) light. Sometimes, though, the term carcinogen is used more narrowly to refer to just those substances (like the constituents of cigarette smoke, for example) that have a well-understood, DNA-damaging mechanism. We'll adopt the broad

Smoking (LO#2, LO#3)

Smoking, particularly cigarette smoking, causes more premature deaths in the United States than any other environmental factor. Many of those premature deaths are due to diseases such as myocardial infarction and stroke, both significantly more common in smokers. But smoking is a major risk factor for many different types of cancer, too—lung cancer, of course, but also cancers of the mouth, esophagus, pancreas, and bladder, many of which have a high mortality rate.

ACTIVE LEARNING QUESTION

Review: Which lung cancer is least associated with cigarette smoking?

A. ○ Lung adenocarcinoma
B. ○ Squamous cell carcinoma of the lung
C. ○ Small cell lung carcinoma

Explanation (requires correct answer)

Diet (LO#2, LO#3)

The list of dietary factors that may increase cancer risk is long, and it seems to be constantly growing and changing. Some of these factors, however, have been well-studied and are widely accepted as definite risk factors. For example, excessive amounts of refined carbohydrates and fat as well as processed red meats are associated with an increased risk of colorectal cancer.

Colonoscopy is recommended as a screening for colon cancer beginning at what age for average-risk patients?

Alcohol Consumption (LO#2, LO#3)

Excessive alcohol consumption increases risk for cancers of the mouth, larynx, and esophagus. If the patient is also a smoker, the risk of these cancers is synergistically increased with alcohol use. Many patients with long-standing alcohol abuse develop alcoholic cirrhosis of the liver, a condition that predisposes the patient to developing hepatocellular carcinoma, or cancer of liver cells. Alcohol al e body, which increases the risk of breast cancer.

Review: Name the carcinogenic alcohol metabolite and its mechanism for carcinogenesis.

Obesity (LO#2, LO#3)

Obesity is increasing at an alarming rate not only in the United States, but also in other parts of the world. In women, obesity leads to excessive estrogen exposure, which increases the risk of breast and endometrial cancer. But obesity also contributes to cancer deaths in men. The increase in cancer risk is not trivial. In fact, the rates of cancer death in patients with morbid obesity are more than 50% higher than they are in the rest of the population.

Reproductive History (LO#2, LO#3)

As mentioned previously, excessive estrogen exposure increases the risk of several types of canc etrium, and ovarian cancer. During pregnancy, higher levels of progesterone counterbalance estrogen, giving a little reprieve from estrogen stimulation. The fewer pregnancies a woman has, and the later in life they occur, the more estrogen she is exposed to overall.

Radiation Exposure (LO#2, LO#3)

Whether from therapeutic interventional procedures (such as radiation treatments for cancer) or from the sun's UV rays, radiation significantly increases the risk of cancer. Both types of radiation may damage the cell's DNA, leading to genetic mutations that can pave the way for the development of cancer.

Review: How does ultraviolet light B (UVB) damage DNA?

Infectious Agents (

Infectious agents cause approximately 15% of all cancers worldwide, with a much higher incidence in developing countries. Several viruses are known to cause cancers (for example, HPV causes cervical cancer, and hepatitis viruses may cause hepatocellular carcinoma). Bacterial agents are less commonly implicated than viral agents, but one well-researched example is the bacterium *Helicobacter pylori*, which is causally related to both gastric carcinoma and a particular type of lymphoma called mucosa-associated lymphoid tissue (MALT) lymphoma. Infectious agents associated with cancer will be discussed in a separate brick.

Occupational Agents (LO#2, LO#3)

Several carcinogens are encountered primarily in occupational settings. A wide range of fields, from agriculture to construction to manufacturing, use specific agents that are known to cause cancer:

- Asbestos (construction materials) is linked to mesothelioma
- Benzene (printing, paint, and light oil) is linked to acute myeloid leukemia.
- Radon (underground mines, some residential properties) is linked to lung cancer.
- Vinyl chloride (refrigerants, adhesives) is linked to hepatic angiosarcoma.

As we can see, carcinogens are both diverse and widespread, and a solid awareness of them is critical to successfully preventing many cancers.

What type of cancer is linked to benzene exposure?

Age (LO#1, LO#2, LO#3)

Age is also an important risk factor for cancer development, and the incidence of cancer varies by age. Most cancers occur in patients over 55 years of age. However, pediatric patients do develop cancer that are often distinct from those encountered in adult populations, most commonly acute leukemias and neoplasms of the central nervous system. Common infant and childhood cancers include the following and will be discussed in more depth in relevant organ system courses.

- Neuroblastoma
- Wilms Tumor
- Retinoblastoma
- Acute lymphoblastic leukemia
- Rhabdomyosarcoma

Review: What important screening is performed for retinoblastoma?

What Are the Genetic Risk Factors for Cancer? (LO#2, LO#3)

Because cancer arises from mutations in DNA, it is not surprising that heredity plays a role in cancer. This brings up the distinction of cancers that are inherited versus those that are sporadic; that is, those arising de novo. Sporadic cancers can be further classified as spontaneous cancers, which arise without carcinogen exposure, and induced cancers, which are the result of carcinogen exposure.

Sporadic cancers do not come from inherited genetic mutations. Instead, at some point in the individual's life, he or she acquires a series of mutations in a somatic cells, which then leads to cancer. The somatic cell line is not passed onto the next generation and is thus considered sporadic. Lung and bladder cancers are examples of sporadic cancers due to carcinogen exposure.

On the other hand, hereditary cancers are caused by gene mutations that are passed on from the p sed some tumor predisposition syndromes, which are caused by inherited mutations of tumor suppressor genes. Let's review them here.

ACTIVE LEARNING QUESTION

Review: Match the tumor predisposition-syndrome related neoplasm with the altered tumor suppressor gene.

RB1	✓ Retinoblastoma
TP53	✓ Breast cancer
APC	✓ Colorectal adenocarcinoma
NF1	✓ Neurofibroma

Explanation

RB1 is mutated in retinoblastoma tumor predisposition syndrome. Breast cancer may arise in the context of germline *TP53* mutations in Li Fraumeni syndrome. Familial adenomatosis polyposis syndrome has a nearly 100% chance of colorectal adenocarcinoma development and is due to a germline mutation in *APC*. Neurofibromatosis 1 is caused by inherited *NF1* mutations and manifests with café-au-lait macules and multiple n

Relevant to today's discussion of breast cancer, germline mutations of *BRCA1* and *BRCA2* genes are well-known risk factors for the development of breast and ovarian carcinomas before the age of 50 years. These genes are important to DNA repair, and their absence creates genomic instability that can lead to cancer. Because patients with *BRCA1/2* mutations are considered high-risk for breast cancer development, many experts recommend the need for magnetic resonance imaging (MRI) screening before the age of 30. Additionally, some patients may choose to undergo prophylactic mastectomy (surgical removal of the breast) to decrease the risk of breast cancer development.

Hereditary cancers tend to manifest earlier and act more aggressively than their sporadic counterparts. For example, colon cancer can be either sporadic or hereditary. Sporadic cases usually appear after age 50 years, and if detected early, typically have a relatively good prognosis. In contrast, hereditary forms of colon cancer, such as those that develop in patients with Lynch syndrome, can occur as early as age 20 years and typically have a poorer prognosis.

CASE CONNECTION BACK TO INTRODUCTION ↑

Returning to the patient with the new mouth ulcer, you reflect that environmental factors and exposures play a major role in cancer development. Upon further questioning, you learn that the patient smokes a pack of cigarettes a day and consumes alcohol fr use are risk factors for squamous cell car edule the patient for a biopsy, and the pathologist confirms the diagnosis of SCC. A definitive surgical resection is then planned to treat the patient's cancer.

Summary

- Earlier detection of cancer has changed cancer mortality and prevalence.
- The most common cancers in men and women are prostate and breast cancer, respectively.
- The cancer that causes the most deaths in both men and women is lung cancer, and many of these cases are preventable by the avoidance of tobacco smoke
- Cervical cancer has decreased in prevalence in the United States largely due to the introduction of the Pap smear.
- Eighty percent of cervical cancers occur outside the United States in developing countries, where access to Pap smears is limited.
- Environmental factors, such as dietary choices and smoking, play a major role in the development of many types of cancer.
- Cancers may be sporadic or hereditary.
- Hereditary cases appear at younger ages and act more aggressively than their sporadic counterparts.

Review Questions

1. Which of the following cancers results in the highest absolute number of deaths of females i

A. ○ Breast cancer
B. ○ Cervical cancer
C. ○ Colorectal cancer
D. ● Lung cancer
E. ○ Pancreatic cancer

Hide Explanation

The correct answer is lung cancer (D). Lung cancer is the number one cause of cancer-related mortality in both males and females. Breast cancer (A) is the most common cancer in females but the second most common cause of cancer mortality in women. Cervical cancer (B) is very rare because of screening and does not make the top-10 cancer-related deaths. Colorectal (C) and pancreatic (E) cancer are the third and fourth most common causes of cancer mortality, respectively, in females.

2. At what age is initiation of screening for colorectal cancer routinely recommended in a healthy individual without a family history of colon cancer?

A. ○ 21 years
B. ○ 30
C. ○ 40 years
D. ● 45 years
E. ○ 65 years

Hide Explanation

The correct answer is 45 years (D). Screening for colorectal cancer in a healthy individual without a family history of colorectal cancer is recommended to begin at age 45 years and repeats every 10 years. Patients with a family history of colorectal adenocarcinoma may begin screening at 40 years (C) or 10 years prior to onset in the family member. Average-risk women often begin pap smear screening at the age of 21 (A). Women with *BRCA1/2* mutations or at high risk for breast cancer may begin screening at age 30 (B) or earlier. Since most cancers occur in patients over 55 years of age, routine screening beginning at the age 65 would not be an effective strategy for cancer prevention (E).

3. A 40-year-old sexually active female who is a construction worker with a 20 pack-year smoking history and a poor diet (consisting of excessive amounts of refined carbohydrates and fat) comes to the emergency department with a 2-month history of unintended weight loss. She is concerned that she has cancer. Which of the following cancers is matched with the correct environmental factor to which this female has been exposed?

A. ○ Breast cancer, refined carbohydrate intake
B. ○ Colorectal cancer, UV exposure
C. ○ Gastric cancer, human papillomavirus infection
D. ○ Mesothelioma, smoking
E. ● Pancreatic cancer, smoking

Hide Explanation

The correct answer is E. Pancreatic cancer is associated with smoking. Breast cancer is not associated with excessive intake of refined carbohydrates (A) but rather estrogen exposure. Colorectal cancer is associated with excessive intake of refined carbohydrates and fat but not UV radiation (B). UV radiation is a risk factor for certain skin cancers, such as melanoma and squamous cell carcinoma. *H pylori* infection can cause gastric cancer, but HPV infection has not been shown to be linked to gastric cancer (C). Mesothelioma is caused by asbestos exposure, not smoking (D), the latter of which is highly associated with squamous cell carcinoma and small cell carcinoma of the lung.

How would you rate this Brick?

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Explain the epidemiology of cancer and how cancer incidence has changed over time~~
02 ☑ ~~Discuss key risk factors for common cancers~~
03 ☑ ~~Formulate strategies to reduce the risk of cancer~~

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Objective Confidence: 3/3